

Date	17 July 2026
Project Name	Smart Lender AI: An Intelligent Machine Learning System for Loan Approval Prediction
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Maximum Marks	3 Marks

PROJECT PLANNING & RESOURCE ESTIMATION

This **Project Planning** document provides the development timeline, phase activities, resource allocations, and effort estimations for the **Smart Lender AI** implementation pipeline [cite: 20]. The timeline is designed to support structured agile sprints, leading to an efficient system rollout [cite: 20].

1. Development Lifecycle Timeline

PHASE	ACTIVITY FOCUS	KEY TASKS & ENGINEERING GOALS	DURATION
Phase 1	Requirements & Architecture	Map user journeys, compile functional and non-functional requirement grids, and design the three-tier system architecture boundaries [cite: 20, 29].	Week 1
Phase 2	Data Engineering & Modeling	Clean structural credit data, handle null properties, apply min-max scaling, train tree classifiers (XGBoost/Random Forest), and serialize pipelines [cite: 20].	Weeks 2-3
Phase 3	API & UI Development	Build the lightweight Flask REST routing endpoints and construct the responsive web form interfaces using Bootstrap 5 [cite: 20, 29].	Weeks 4-5
Phase 4	Integration & Testing	Connect the application layers, execute verification suites, validate sub-second inference speed bounds, and perform dashboard accuracy reviews [cite: 20].	Week 6
Phase 5	Deployment & Review	Deploy the web application into sandboxed container instances and wrap up final technical documentations [cite: 20].	Week 7

2. Resource Allocation & Effort Breakdown

The total engineering investment spans **7 Weeks**, organized into clear resource domains to keep execution balanced:

RESOURCE DOMAIN	PRIMARY DELIVERABLES	ESTIMATED EFFORT
Data Engineering Core	Statistical matrix cleansing, feature scaling configuration records, and model training optimizations [cite: 20].	30% Total Effort
Backend System Engineering	Flask routing components, request validation routines, and in-memory data pipeline routing execution [cite: 20, 29].	35% Total Effort
Frontend UI/UX Design	Responsive application forms and credit score explanation dashboards [cite: 20, 29].	20% Total Effort
Quality Assurance & Testing	Stress checking routing endpoints, handling edge case constraints, and verifying response latencies [cite: 20].	15% Total Effort

Project Planning Strategy:

This development plan runs UI building and ML model generation in parallel, minimizing cross-team blockers. Dedicating the final stages to integration verification ensures the Flask middleware meets the target sub-second response times prior to release [cite: 20].